CSTA Canterbury Science Teachers Association	**Student Name: Kayson Lee** Teacher code: WARD

	Elim C	

Science Practice Examination 2025

NCEA Level 1

Achievement Standard 92022

Chemistry and Biology 1.3

Demonstrate understanding of genetic variation in relation to an identified characteristic

Credits: 5

Suggested Time: 60 minutes

Instructions

Answer all questions in the spaces provided.

You must hand this examination paper to the supervisor at the end of the examination.

Check that this paper has all 8 pages numbered and in the correct order.

Achievement Criteria. (for assessors only)		
Achievement	Achievement with Merit	Achievement with Excellence

Demonstrate understanding of genetic variation in relation to an identified characteristic		Explain understanding of genetic variation in relation to an identified characteristic		Evaluate understanding of genetic variation in relation to an identified characteristic	
			Overall Level of performance:		

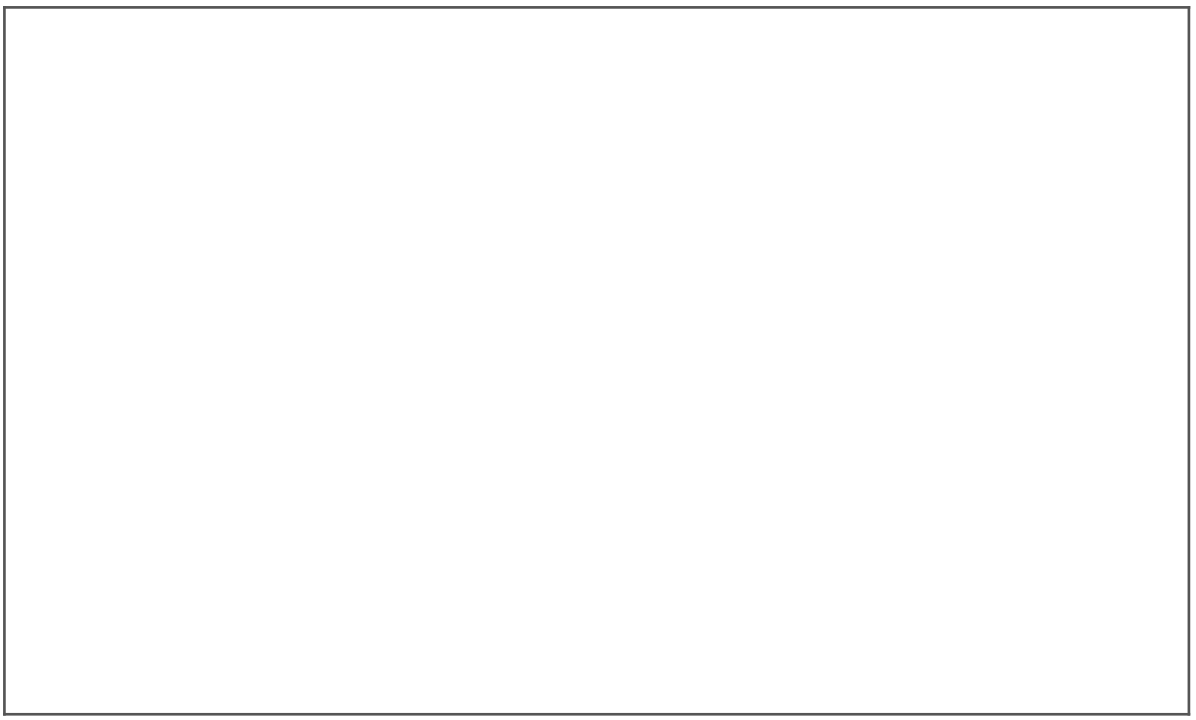
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Question one – Leucism in Weka

Weka are normally a plain brown, flightless bird found throughout the South Island of New Zealand. *Leucism* is a genetic condition that causes partial or total loss of pigmentation in animals. The result is usually a white or pale individual.

Explain how leucism can occur and evaluate how this trait could potentially be passed down to future generations.

State definitions for DNA, genes, alleles and chromosomes, and explain the relationships between them. You may like to include a labelled diagram to help with your explanation



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Define the term mutation, and explain how mutations can occur

Evaluate how the white trait could be passed on to future generations of weka

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Question Two – Natural selection in Anna's hummingbird

Anna's hummingbird (*Calypte anna*) is a North American species of hummingbird. Over the past few decades researchers have noticed a rapid change in beak shape, which has been linked to the use of artificial bird feeders in people's gardens. Their beaks have become longer and larger, which helps to gather more of the sugar water in the bird feeder.

Male hummingbirds have adapted to having pointier and sharper beaks than normal, which helps them to fight other males in order to compete for the sugar water in the feeders.

Explain how the bird feeders have artificially affected the evolution of Anna's hummingbirds.

Define the terms evolution and natural selection

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Describe where the natural variation in beak size originates

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Explain how the bird feeders have driven the change in beak shapes and sizes

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Evaluate in evolutionary terms, how the changing beak shapes and sizes have made Anna’s hummingbirds successful in their current environment

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Question Three – Huntaway dogs and gene tracking

The Huntaway is the first dog breed that is recognised to have originated in New Zealand. Their desirable characteristics include their energy and endurance, loud bark for herding sheep and intelligence and trainability.

Some Huntaway dogs carry recessive alleles that can cause diseases that affect their hearts and joints, making them unsuitable for work on the farm.

Geneticists at Massey University have identified genetic markers that allow breeders to identify these recessive traits before they are inherited.

Explain the importance of using genetic markers to identify recessive disease-causing traits in Huntaway dogs.

Define recessive traits and genetic markers

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Explain why some individuals (called **carriers**) can carry recessive alleles but not be affected by them, and explain how recessive traits are expressed in individuals

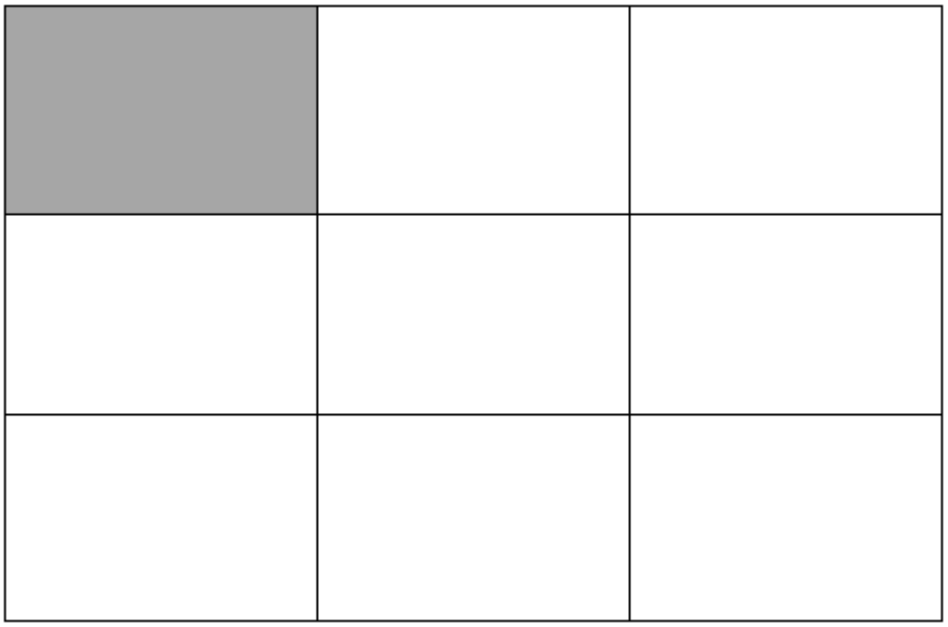
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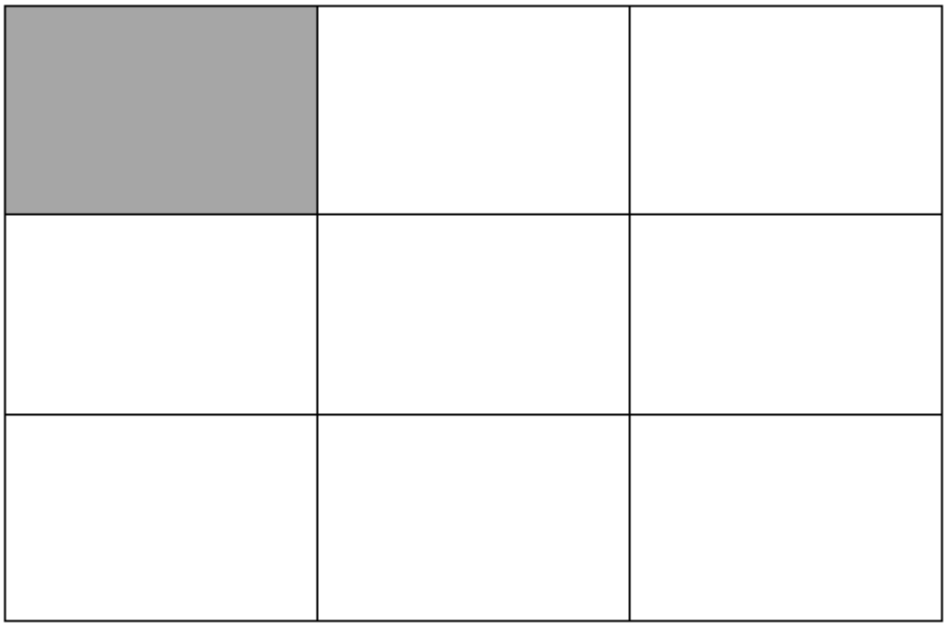
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Punnett squares (if needed for your explanation)





Explain why it is hard to identify if a Huntaway dog is a carrier of a recessive trait

Evaluate how using genetic markers can help dog breeders avoid breeding dogs that carry recessive traits, therefore increasing the overall health of the Huntaway dog breed

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